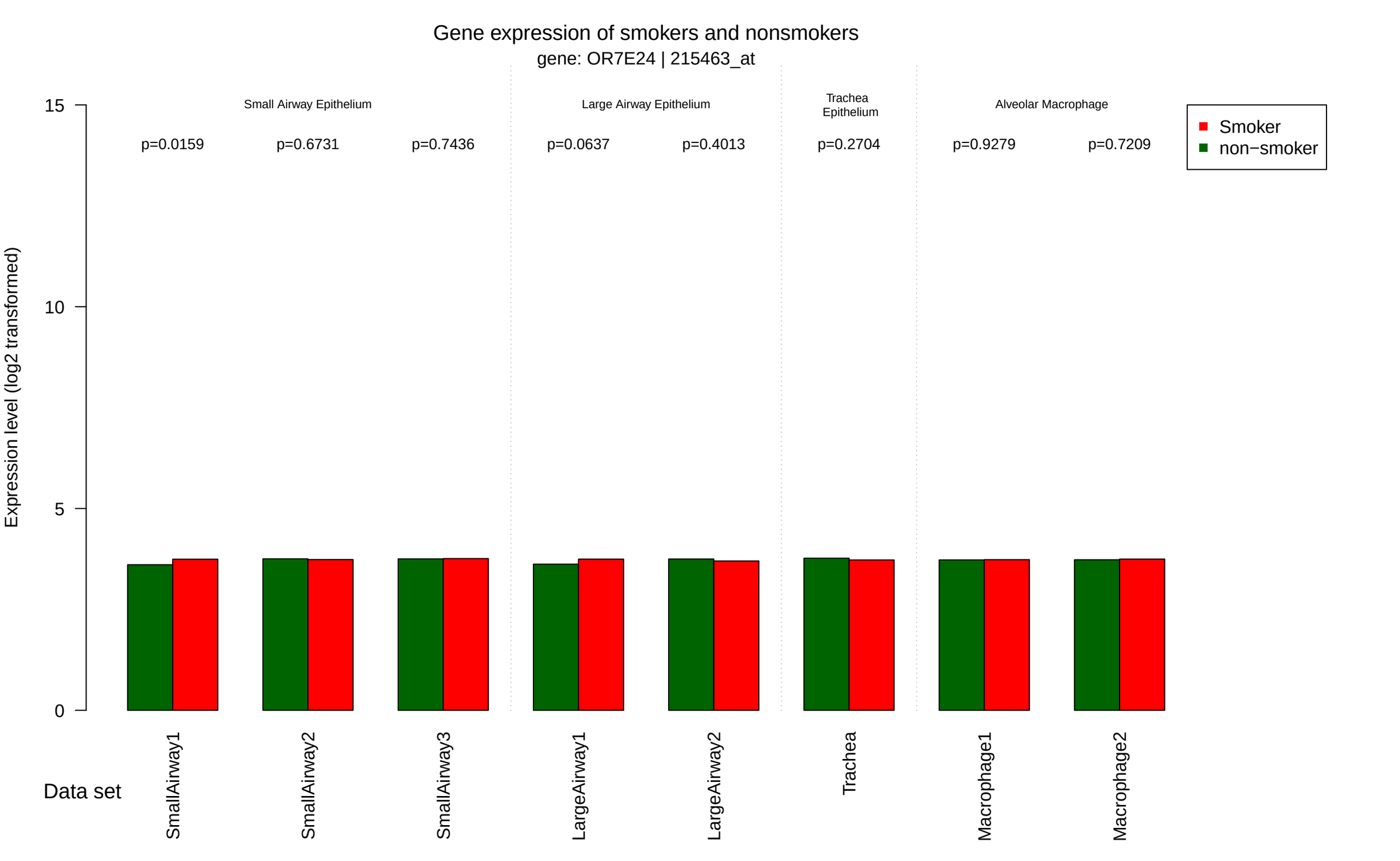
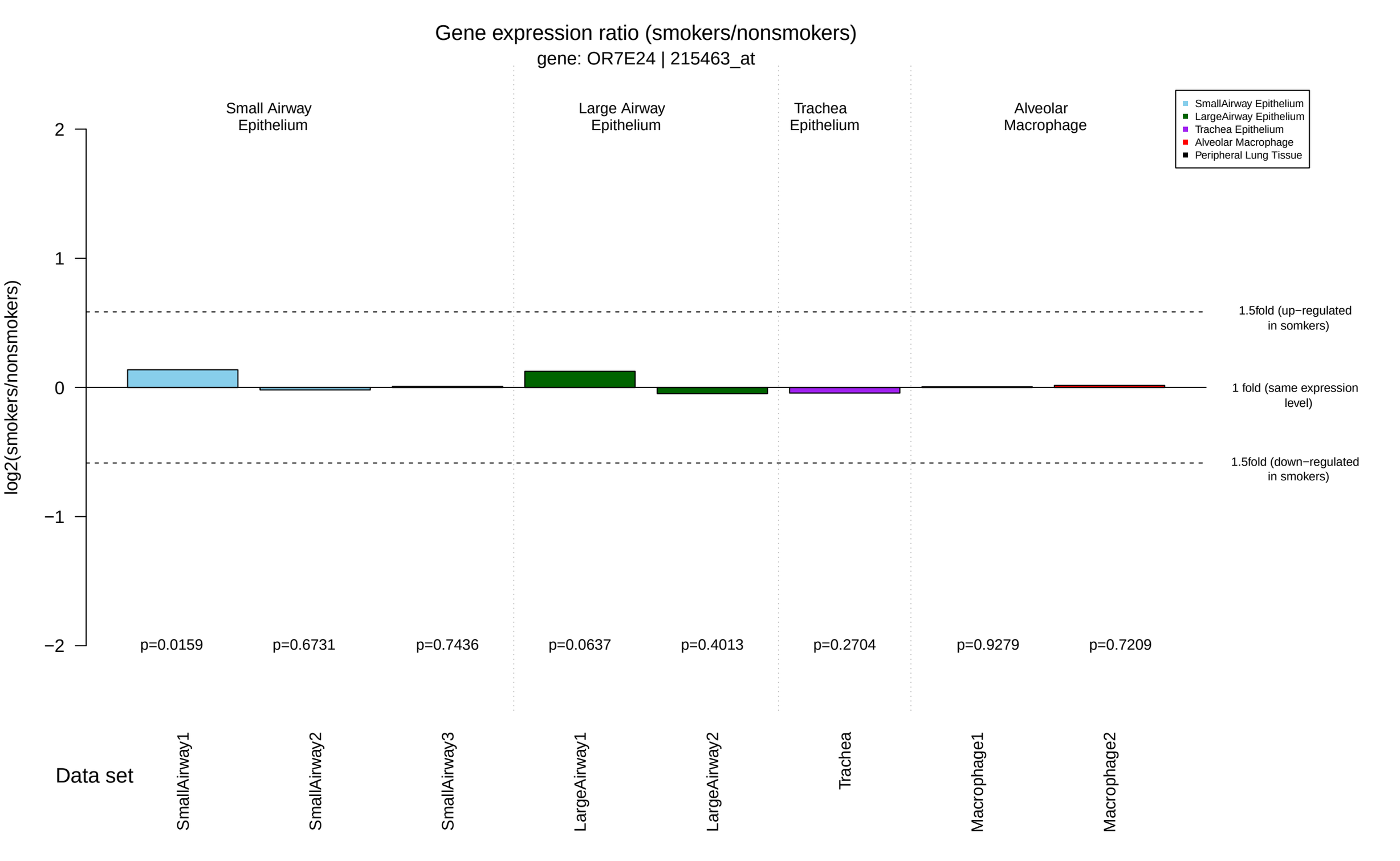
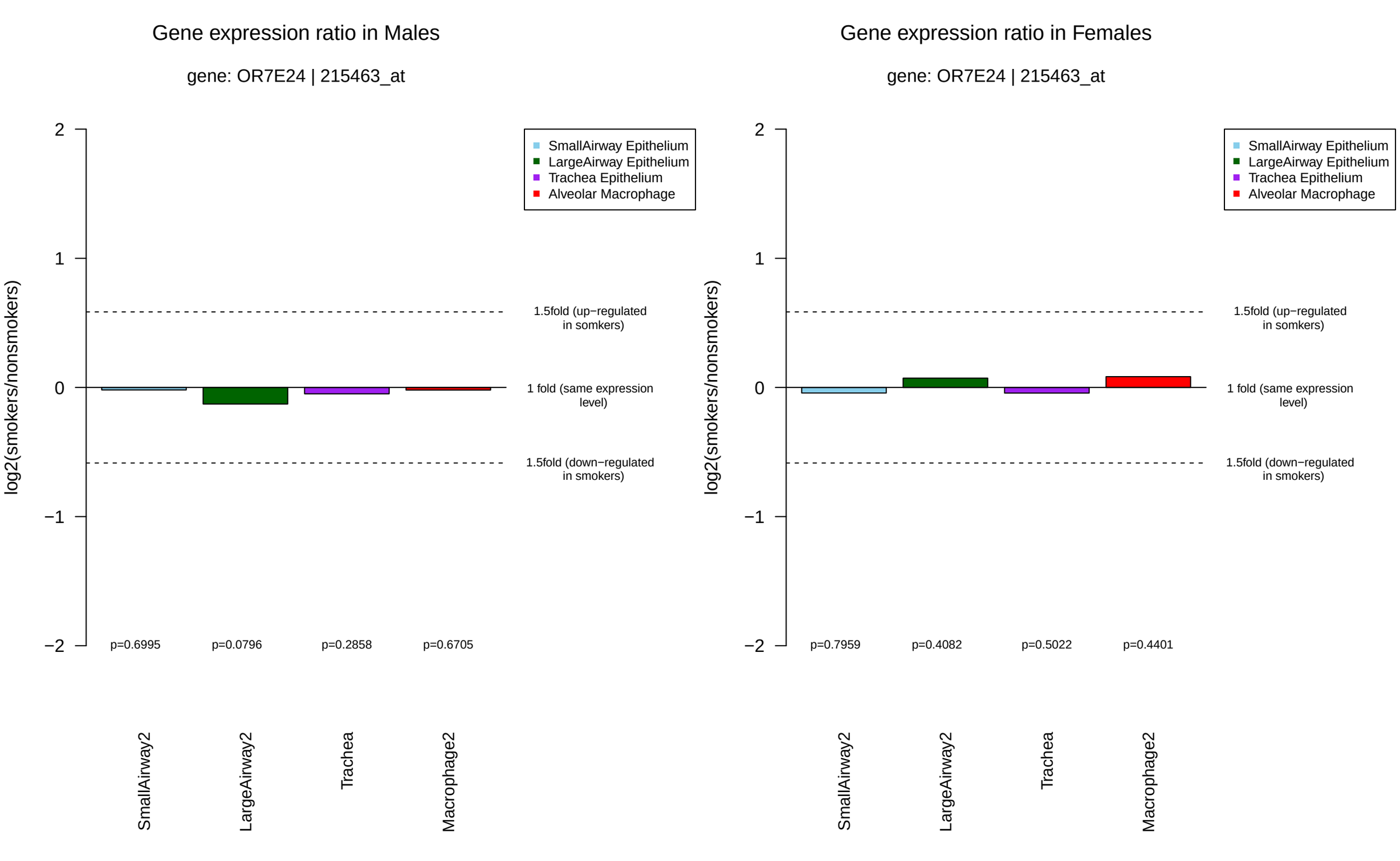
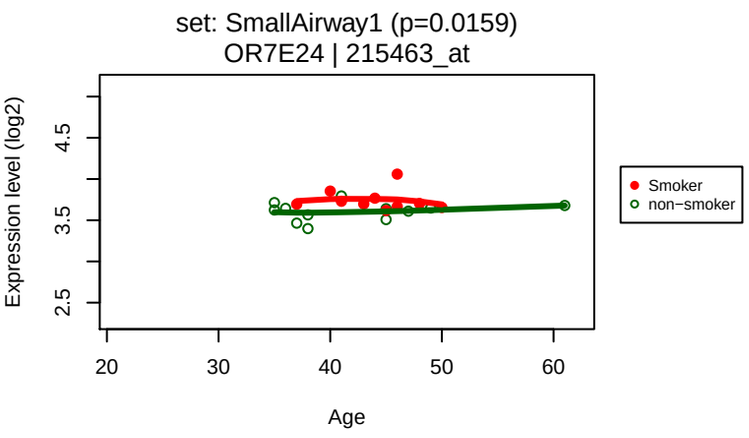
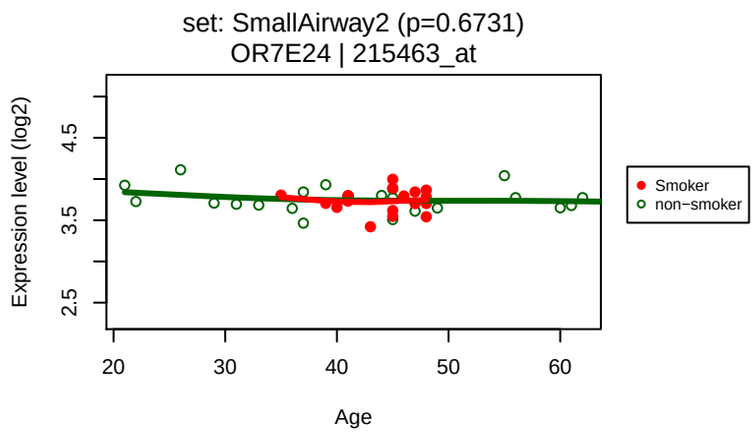
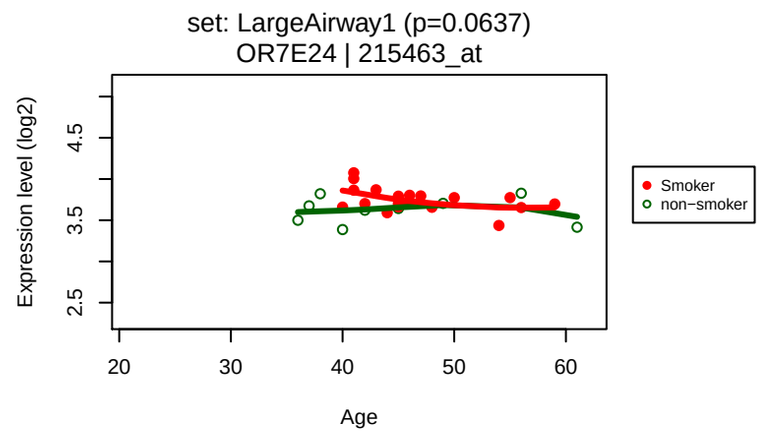

Gene expression of smokers and nonsmokers
gene: OR7E24 | 215463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0159
p=0.6731
p=0.7436
p=0.0637
p=0.4013
p=0.2704
p=0.9279
p=0.7209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

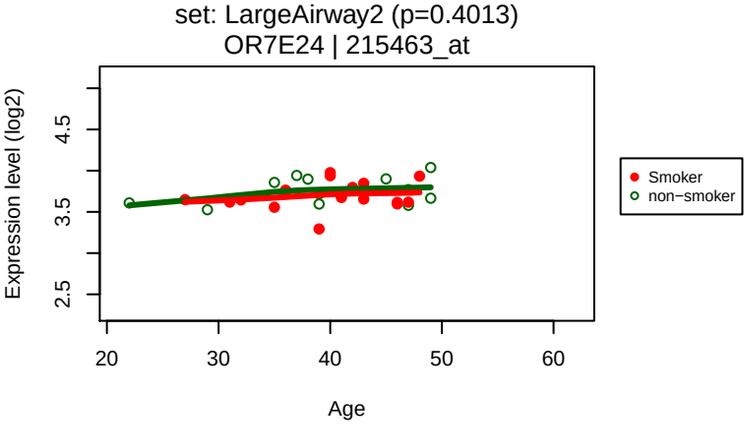

Gene expression ratio (smokers/nonsmokers)
gene: OR7E24 | 215463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0159
p=0.6731
p=0.7436
p=0.0637
p=0.4013
p=0.2704
p=0.9279
p=0.7209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

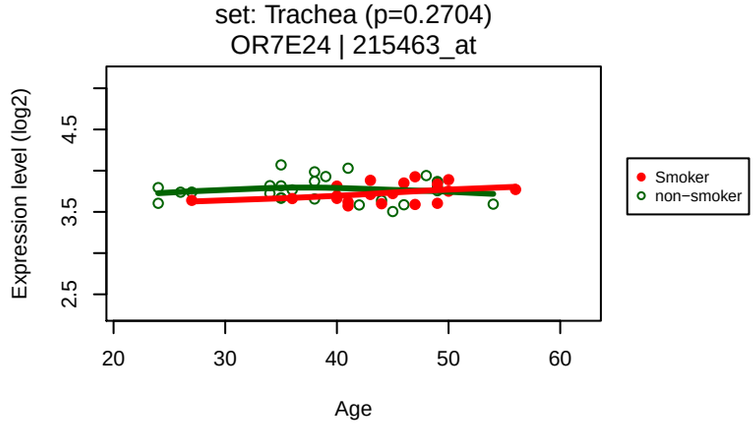

Gene expression ratio in Males
gene: OR7E24 | 215463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6995
p=0.0796
p=0.2858
p=0.6705
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR7E24 | 215463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7959
p=0.4082
p=0.5022
p=0.4401
SmallAirway2
LargeAirway2
Trachea
Macrophage2

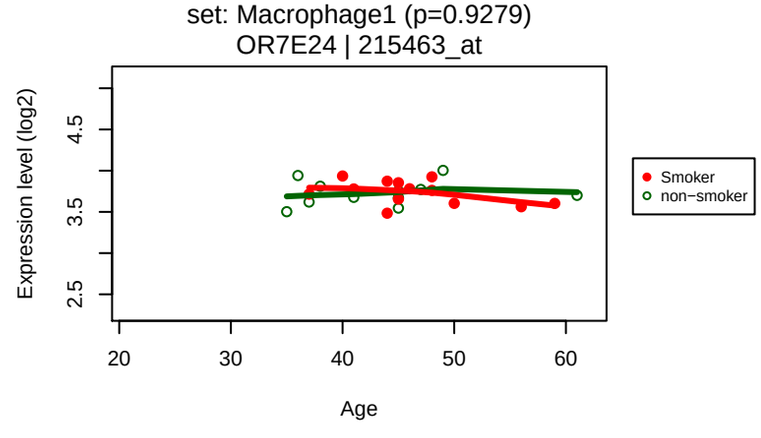

set: SmallAirway1 (p=0.0159)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker

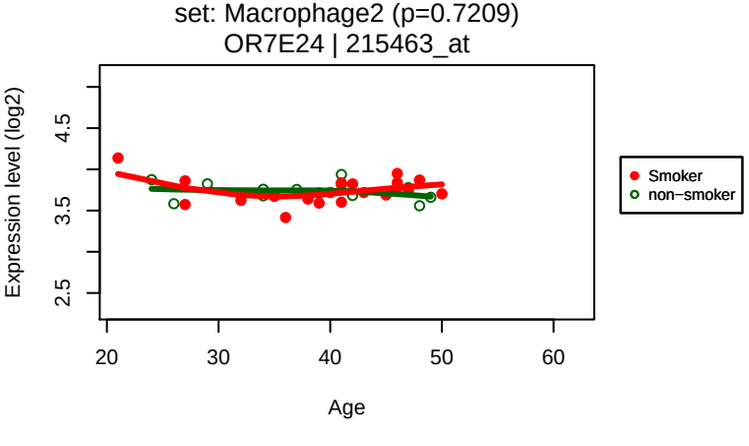

set: SmallAirway2 (p=0.6731)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker

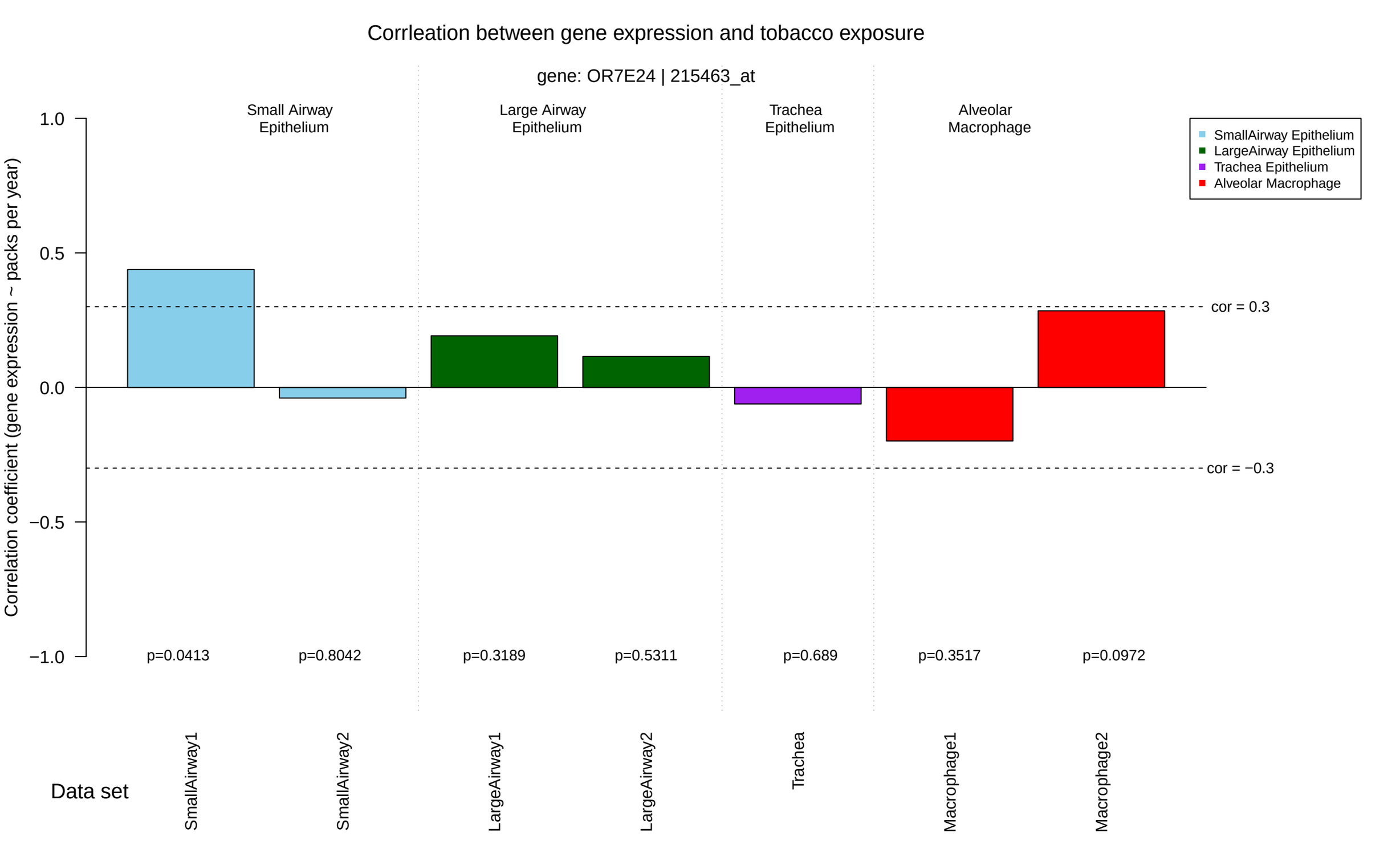

set: LargeAirway1 (p=0.0637)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4013)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2704)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9279)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7209)
OR7E24 | 215463_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR7E24 | 215463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0413
p=0.8042
p=0.3189
p=0.5311
p=0.689
p=0.3517
p=0.0972
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2